STUDENT RECORD SYSTEM

CODE COVERAGE REPORT

Statement and interpretation of executed source-line coverage

Document	TEST-COV-03
Project	Student Record System
Course	Software Engineering Mini Project
Team	Team 11
Implementation	C++17 application with SQLite persistence
Report basis	Recorded project test results and coverage.xml evidence available in the project conversation
Report date	10 November 2026

Prepared as a final testing report for the project documentation package.

1. Objective

This report presents source-line coverage from the latest supplied GCOVR 7.2 coverage.xml artifact. Line coverage measures the proportion of instrumented executable source lines that were executed by the test workload.

2. Measurement Tools and Evidence

Item	Recorded value
Instrumentation	GCC coverage instrumentation (--coverage)
Report generator	GCOVR 7.2
Evidence file	coverage.xml supplied from the GitHub Actions coverage artifact
Coverage package	src
Coverage timestamp in XML	1794292553 (Unix timestamp recorded in the XML)

3. Overall Line Coverage

69.58%	803 / 1,154	351
Line coverage	Lines executed / valid	Lines not executed

Calculation: 803 covered lines ÷ 1,154 valid lines × 100 = 69.584%, reported as 69.58%.

4. Interpretation

• 803 instrumented source lines were executed by the coverage workload.

• 351 valid source lines were not executed by that workload.

• The result reflects the source files included in the configured coverage report and exercised by the associated tests.

• A line not executed during testing is not necessarily defective; it identifies code paths not reached by the measured test workload.

5. Coverage Scope and Limitations

• The supplied XML identifies the coverage package as src.

• The report is based on the supplied coverage.xml artifact, not a fresh local rerun.

• Coverage percentages depend on the source files included by the workflow's GCOVR filters and the tests executed in that run.

• Line coverage does not establish correctness, requirement completeness, or absence of defects.

• The XML is the source of truth for the aggregate line counts in this report.

6. Result

The supplied coverage artifact reports 69.58% line coverage (803 of 1,154 valid lines). This is the measured line-coverage result for the recorded coverage run.

7. Conclusion

The coverage workload executed approximately 69.58% of the valid source lines included in the report. The remaining 30.42% were not executed by that workload and represent opportunities for additional test scenarios if greater line coverage is desired.